

1 Spin

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Date: June 28, 2026

Spin is some internal property of particles, like mass. To get the value of it, we need a tool that is called spin operator $\mathbf{S}$. The spin operator is a vector and the basis of this operator space is Pauli matrix. The explicit expression for spin operator is:

$$\hat{\mathbf{S}} = \frac{\hbar}{2} \boldsymbol{\sigma} \quad (1.1)$$

The detailed expression for Pauli matrix is:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (1.2)$$

Note here the representation of the Pauli matrix is based on the basis $|\uparrow\rangle$ and $|\downarrow\rangle$, which means the up-spin and down-spin of electrons along the z-axis.

1.1 Rotation of spin

The expression for the rotation of quantum state is that:

$$R(\theta, \hat{\mathbf{n}}) = \exp\left(-\frac{i}{\hbar} \theta \hat{\mathbf{n}} \cdot \hat{\mathbf{J}}\right) \quad (1.3)$$

Where $\hat{\mathbf{n}}$ is the axis of the rotation, and the θ is the angle. Considering the Lie algebra of the Pauli matrix has the same structure as the rotation, so we can define the rotation of spin as:

$$D(\theta, \hat{\mathbf{n}}) = \exp\left(-\frac{i}{2} \theta \hat{\mathbf{n}} \cdot \boldsymbol{\sigma}\right) \quad (1.4)$$

For the expression, we can expand it in Taylor series, which gives:

$$D(\theta, \hat{\mathbf{n}}) = I + \left(-\frac{i}{2} \theta \hat{\mathbf{n}} \cdot \boldsymbol{\sigma}\right) + \frac{\left(-\frac{i}{2} \theta \hat{\mathbf{n}} \cdot \boldsymbol{\sigma}\right)^2}{2!} + \frac{\left(-\frac{i}{2} \theta \hat{\mathbf{n}} \cdot \boldsymbol{\sigma}\right)^3}{3!} + \dots \quad (1.5)$$

It can be noted that:

$$(\hat{\mathbf{n}} \cdot \boldsymbol{\sigma})^2 = I \quad (1.6)$$

So the formulation can be rewritten as:

$$\begin{aligned} D(\theta, \hat{\mathbf{n}}) &= \left(\sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} \left(\frac{\theta}{2}\right)^{2m}\right) I - i \left(\sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} \left(\frac{\theta}{2}\right)^{2m+1}\right) (\hat{\mathbf{n}} \cdot \boldsymbol{\sigma}) \\ &= \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) (\hat{\mathbf{n}} \cdot \boldsymbol{\sigma}) \end{aligned} \quad (1.7)$$

We may observe that the existence of $\frac{\theta}{2}$, which means the rotation of spin is not SO(3) symmetry. It will not return to the original point when you rotate it as 2π . Instead, you have to rotate it to 4π as a periodic.

1.2 Expression for the eigenstates of spin in any direction

To get the eigenstates of the spin in any direction, it's intuitively that we may think that, under a rotation along the axis which has the same direction, the spin should not change. So the up-spin along that axis should have the equation:

$$\frac{1}{2} \hat{\mathbf{n}} \cdot \boldsymbol{\sigma} |\psi\rangle = \frac{1}{2} |\psi\rangle \quad (1.8)$$

The $\hat{\mathbf{n}}$ vector can be written as: $\hat{\mathbf{n}} = \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix}$. And we have:

$$\begin{aligned} \hat{\mathbf{n}} \cdot \boldsymbol{\sigma} &= \begin{pmatrix} \cos \theta & \sin \theta \cos \phi - i \sin \theta \sin \phi \\ \sin \theta \cos \phi + i \sin \theta \sin \phi & -\cos \theta \end{pmatrix} \\ &= \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix} \end{aligned} \quad (1.9)$$

For any arbitrary states, we can denote it as $\begin{pmatrix} u \\ v \end{pmatrix}$. So we have:

$$\begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u \\ v \end{pmatrix} \quad (1.10)$$

And we can get a relation between u and v :

$$\begin{aligned} u \cos \theta + v \sin \theta e^{-i\phi} &= u \\ v(2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}) e^{-i\phi} &= 2u \sin^2 \frac{\theta}{2} \\ \frac{u}{v} &= \frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2} e^{i\phi}} \end{aligned} \quad (1.11)$$

Thus the eigenstates of the spin in any direction can be expressed as $\begin{pmatrix} e^{-\frac{i\phi}{2}} \cos \frac{\theta}{2} \\ e^{\frac{i\phi}{2}} \sin \frac{\theta}{2} \end{pmatrix}$.

A trick for the calculation is that, $\langle \hat{\mathbf{n}} | \sigma_z | \hat{\mathbf{n}} \rangle = \cos \theta$, where θ is the angle between the direction vector and the z axis. Same for σ_x and σ_y , as $\langle \sigma_x \rangle = \sin \theta \cos \phi$, $\langle \sigma_y \rangle = \sin \theta \sin \phi$.

1.3 Zeeman effect

Zeeman effect is that, the magnetic moment will couple with the external magnetic field $\mathbf{B}$, which gives rise to the energy difference. The Hamiltonian for this effect is:

$$\bar{H} = -\boldsymbol{\mu} \cdot \mathbf{B} \quad (1.12)$$

The magnetic moment usually has the structure as:

$$\boldsymbol{\mu} = (\text{factor}) \times (\text{g factor}) \times (\text{angular momentum}) \quad (1.13)$$

Some typical example is:

- Angular orbital momentum: $\boldsymbol{\mu}_L = -\frac{\mu_B}{\hbar} \mathbf{L}, g \approx 1$
- Spin magnetic momentum: $\boldsymbol{\mu}_S = -2\frac{\mu_B}{\hbar} \mathbf{S}, g \approx 2$
- Total angular momentum: $\boldsymbol{\mu}_J = -g_J \frac{\mu_B}{\hbar} \mathbf{J}$

1.4 Spin orbital coupling

In crystal configuration, it may have the expression as:

$$\bar{H} = g\sigma^i \bar{p}_i \quad (1.14)$$

Which is contributed by the relativity effect, where the electrons couple with the magnetic field generated by it self. So there will have the momentum component in the expression.

1.5 Larmor precession

Here we choose the up spin along z axis as $|\uparrow\rangle$, and the magnetic field is $\mathbf{B} = B\hat{\mathbf{z}}$, and set $\hbar = 1$. The eigenstate along $\hat{\mathbf{n}}$ can be written as:

$$|\hat{\mathbf{n}}\rangle = e^{-\frac{\phi}{2}i} \cos \frac{\theta}{2} |\uparrow\rangle + e^{\frac{i\phi}{2}} \sin \frac{\theta}{2} |\downarrow\rangle \quad (1.15)$$

So the Lagrange of the system can be written as:

$$L = \langle \hat{\mathbf{n}} | i \frac{d}{dt} | \hat{\mathbf{n}} \rangle - \langle \hat{\mathbf{n}} | \hat{H} | \hat{\mathbf{n}} \rangle \quad (1.16)$$

The first term is:

$$\langle \hat{\mathbf{n}} | i \frac{d}{dt} | \hat{\mathbf{n}} \rangle = \frac{1}{2} \cos \theta \dot{\phi} \quad (1.17)$$

And the second term is:

$$\langle \hat{\mathbf{n}} | \hat{H} | \hat{\mathbf{n}} \rangle = \langle \hat{\mathbf{n}} | -\mathbf{B} \cdot \hat{\mathbf{S}} | \hat{\mathbf{n}} \rangle = -\frac{1}{2} B \cos \theta \quad (1.18)$$

The Lagrange is:

$$L = \frac{1}{2} \cos \theta \dot{\phi} + \frac{1}{2} B \cos \theta \quad (1.19)$$

And we can get the equation of motion:

$$\frac{\partial L}{\partial \theta} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = -\frac{1}{2} B \sin \theta - \frac{1}{2} \dot{\phi} \sin \theta = 0 \quad (1.20)$$

$$\frac{\partial L}{\partial \phi} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = \frac{1}{2} \sin \theta \dot{\theta} = 0 \quad (1.21)$$

Which gives that:

$$\dot{\phi} = -B \quad (1.22)$$

$$\dot{\theta} = 0 \quad (1.23)$$

It's exactly $\frac{d}{dt} \hat{\mathbf{n}} = \hat{\mathbf{n}} \times \mathbf{B}$. It's so called Larmor precession.